

# One optical map, different simulation budgets

## Binomial GP and RESuM

Predict population-level detection probability throughout the liquid-argon volume.

Which statistical model best uses sparse simulation counts?

# The data are counts, not just noisy fractions

| Per simulation  | Meaning                         | Role in the count GP       |
|-----------------|---------------------------------|----------------------------|
| $s = (x, y, z)$ | Physical voxel center           | Spatial input              |
| $m$             | Primaries with at least one hit | Integer observation        |
| $N$             | Total simulated primaries       | Known likelihood parameter |
| $y = m / N$     | Observed detection fraction     | Reporting and validation   |

Same physics + same primary-generation distribution  $\rightarrow$  one shared  $p(s)$ .

# The binomial GP learns a shared probability field

## Spatial model

$$f(s) \sim \text{GP}(0, k)$$

$$p(s) = 1 / [1 + \exp(-f(s))]$$

Probabilities stay between 0 and 1.

## Count observations

$$m_i \mid f(s_i), N_i \sim \text{Binomial}(N_i, p(s_i))$$

Different  $N$ , same probability function.

Zero counts enter without pseudocounts.

Laplace inference • spatial kernel + constant component • training-only coordinate scaling

# The primary budget determines counting precision

**Example: a voxel with  $p = 0.003$**

| Primaries N | Expected detections | SD of observed fraction | P(zero detections) |
|-------------|---------------------|-------------------------|--------------------|
| 500         | 1.5                 | 0.00245                 | 0.223              |
| 1500        | 4.5                 | 0.00141                 | 0.011              |
| 5000        | 15.0                | 0.00077                 | 2.99e-07           |

A zero-hit file is evidence for a low probability; it does not imply  $p = 0$ .

The likelihood accounts for N directly. No class balancing is required for this objective.

# RESuM first denoises events, then learns the map

## 1 Event representation

Primary features + labels  
Context / target episodes  
Class-aware mixup



## 2 CNP denoising

Predict event scores  
Average scores per voxel  
LF and HF summaries



## 3 Spatial MFGP

LF CNP means  
HF CNP means  
HF raw target fractions

Current GP stage: Gaussian likelihoods in log space, with learned noise per output level.

# The main difference is the observation model

| Feature                 | Current RESuM                     | Binomial GP                       |
|-------------------------|-----------------------------------|-----------------------------------|
| Learning input          | Event features → voxel summaries  | Voxel centers + integer counts    |
| Spatial outputs         | Three related GP output levels    | One probability function          |
| Observation model       | Gaussian on transformed summaries | Binomial on $m$ given $N$ and $p$ |
| Counting precision      | Not explicitly set by each $N$    | Determined by $p$ and each $N$    |
| Zero-hit handling       | Log-space smoothing / floors      | Directly supported                |
| Event-level predictions | Available from the CNP            | Not provided                      |

# Counting noise belongs in training and prediction

## Original RESuM prediction

Gaussian noise learned in log space.

Its scale may absorb counting noise and model mismatch.

Produces lognormal response intervals.

## RESuM diagnostic test

Draw latent GP probabilities, then draw binomial counts.

Changes the predictive model; training still used Gaussian noise.

## Binomial GP

Use the count likelihood in fitting and for new simulations.

Prediction combines latent uncertainty with binomial counting noise.

At a fixed voxel:  $\text{Var}(R \mid s, D) = \text{Var}(P \mid s, D) + E[P(1 - P) \mid s, D] / N$

# Strengths and limitations depend on the task

## RESuM

- + Uses event features and context
- + Can learn transferable denoising
- + Supports related fidelity outputs
- More training and modeling choices
- Mixup may bias mean scores
- CNP uncertainty not fully propagated

## Binomial GP

- + Direct link from counts to probability
- + Handles zero hits and variable N
- + Fewer stages; easier to audit
- Discards event-level features
- Binomial assumptions may fail
- Approximate posterior; dense GP cost

# Current binomial-GP validation: 55 HF voxels

**Bias +0.000085 MAE 0.000680 RMSE 0.001091 Pearson r 0.849**

Coverage: 72.7% / 92.7% / 98.2%. A 55-voxel sample cannot establish 99.7% calibration.

# The saved runs are not a controlled comparison

| Dataset / evaluation | Saved RESuM run            | Saved binomial-GP run                 |
|----------------------|----------------------------|---------------------------------------|
| LF training          | 159 nonzero N1500 voxels   | 760 across N500–N1500; zeros included |
| HF training          | 10 nonzero voxels          | 10 voxels; 1 zero-hit                 |
| HF validation        | 46 nonzero voxels          | 55 voxels; 8 zero-hit                 |
| HF target budget     | 4936 after context split   | Full 5000 primaries                   |
| Interpretation       | Historical workflow result | New dataset and likelihood            |

Use identical files, held-out locations and observed counts before attributing gains to a model.

# Two projections answer different questions

## Marginalized detection probability

Average over omitted coordinates.

$$\bar{p}(x) = E_{Y,Z}[p(x, Y, Z)]$$

Bands quantify uncertainty in an average.

Joint spatial draws are essential.

## Projected detection fractions

Sample an omitted-coordinate location.

Draw its probability, then  $M \sim \text{Bin}(N, P)$ .

Bands include spatial variation, GP uncertainty and counting noise.

A point may miss its exact-coordinate interval yet lie inside the broader projected interval.

# Validation should test accuracy, uncertainty and space

## Mean prediction

Bias: systematic shift  
MAE / RMSE: absolute error  
Pearson / Spearman: trend

## Predictive distribution

Coverage + interval width  
CRPS / weighted interval score  
Likelihood with matching measure

## Spatial generalization

Hold out spatial blocks  
Include zero-hit voxels  
Check distance to training data

Use representative validation; neither approach can recover missing physics from a few widely separated voxels without assumptions.

# Use the binomial GP as the count-level baseline

- Same physics, different N: start with a shared probability field.
- Keep RESuM when event features or transferable denoising add value.
- Compare on matched spatial splits and the same observed counts.
- Use a compact, densely sampled volume to test local interpolation.

Deferred extension: Gaussian likelihood for CNP summaries + binomial likelihood for raw HF counts.

# Sources and implementation provenance

## Method references

Garnelo et al. (2018), Conditional Neural Processes  
<https://proceedings.mlr.press/v80/garnelo18a.html>

Rasmussen & Williams (2006), GPML, Chapter 3  
<https://gaussianprocess.org/gpml/chapters/RW3.pdf>

## Implementation and saved evidence

RESUM\_FLEX: core/binomial\_gp.py; core/surrogates/mfgp\_stage.py

Count-GP run: 20261001T173754\_521023Z

RESuM run: 20260930T195141\_282030Z

Run configurations, counts, metrics and model metadata are the data sources.